| | Liver | Testis | Bone marrow | Kidney | Islet | Eye | Cortex | Lung | $q$ | $S_g$ | profile correlation | π | dN/dS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Recurrent, above the prior** | | | | | | | | | | | | | |
| Receptors (class) | +7.1* | +7.3* | +9.5* | +15.0* | +13.6* | +7.2* | +13.1* | +14.1* | <.001 | 8↑0↓ | 7↑0↓ | −2.2 | +0.9 |
| Epithelial Mesenchymal Transition | +7.8* | +7.6* | +4.0* | +12.1* | +16.7* | +10.0* | +8.1* | +11.6* | <.001 | 8↑0↓ | 6↑0↓ | +0.4 | 0.0 |
| Cytoskeleton & ECM (category) | +5.4* | +10.1* | +5.5* | +8.5* | +10.1* | +7.8* | +3.3* | +9.5* | <.001 | 8↑0↓ | 6↑1↓ | +2.8 | −2.9 |
| KRAS Signaling Up | +2.9* | +5.1* | +5.2* | +6.5* | +9.8* | +6.1* | +4.5* | +8.9* | <.001 | 6↑0↓ | 5↑0↓ | −2.7 | +1.5 |
| Inflammatory Response | +3.7* | +3.4* | +6.4* | +6.3* | +9.9* | +4.0* | +6.1* | +9.1* | <.001 | 7↑0↓ | 6↑0↓ | −4.2 | +3.4 |
| Apical Junction | +4.0* | +2.6* | +3.6* | +7.1* | +9.2* | +6.4* | +5.0* | +7.6* | <.001 | 8↑0↓ | 6↑0↓ | +2.0 | −3.2 |
| TNF-alpha Signaling via NF-kB | +1.3 | +3.7* | +5.0* | +6.7* | +11.1* | +2.8* | +5.0* | +9.6* | <.001 | 6↑0↓ | 6↑0↓ | −1.4 | +1.5 |
| GPCRs (class) | +3.7* | +4.5* | +1.9 | +7.6* | +8.0* | +3.9* | +7.5* | +5.9* | <.001 | 7↑0↓ | 6↑0↓ | −0.1 | −0.5 |
| Coagulation | +4.7* | +4.5* | +2.5* | +6.7* | +8.2* | +4.2* | +4.7* | +6.5* | <.001 | 7↑0↓ | 6↑0↓ | −1.8 | +1.4 |
| IL-2/STAT5 Signaling | +2.7* | +3.2* | +5.1* | +5.5* | +7.4* | +5.6* | +4.1* | +7.3* | <.001 | 7↑0↓ | 5↑0↓ | −1.5 | +1.0 |
| Myogenesis | +4.1* | +4.3* | +2.3* | +6.2* | +7.4* | +5.5* | +4.8* | +6.3* | <.001 | 8↑0↓ | 6↑0↓ | +0.5 | −1.2 |
| UV Response Dn | +3.0* | +4.1* | +1.3 | +5.2* | +8.5* | +5.3* | +5.9* | +5.3* | <.001 | 6↑0↓ | 7↑0↓ | +2.4 | −1.9 |
| Complement | +4.5* | +1.4 | +4.8* | +5.5* | +7.5* | +3.8* | +4.5* | +6.3* | <.001 | 7↑0↓ | 3↑0↓ | −0.9 | +0.6 |
| Hypoxia | +3.0* | +2.1 | +3.8* | +6.1* | +8.0* | +3.9* | +3.5* | +6.3* | <.001 | 6↑0↓ | 4↑0↓ | +0.7 | −1.3 |
| Estrogen Response Early | −0.5 | +1.8 | +2.3* | +7.7* | +7.5* | +5.1* | +4.2* | +7.1* | <.001 | 5↑0↓ | 6↑0↓ | 0.0 | −0.3 |
| Allograft Rejection | +2.6* | +1.3 | +6.7* | +4.5* | +6.3* | +2.9* | +2.2* | +7.0* | <.001 | 5↑0↓ | 7↑0↓ | −3.0 | +2.6 |
| Angiogenesis | | +3.1* | +3.5* | +5.1* | +6.1* | +2.6* | +4.4* | +6.3* | <.001 | 7↑0↓ | 4↑0↓ | −0.1 | +0.1 |
| Signaling pathways (category) | +4.8* | +4.4* | +1.1 | +2.7* | +5.9* | +4.4* | +2.4* | +7.4* | <.001 | 5↑0↓ | 6↑0↓ | +6.4 | −7.7 |
| Ion channels (class) | −0.7 | +7.0* | 0.0 | +5.9* | +4.9* | +2.6* | +9.4* | +3.2* | <.001 | 4↑0↓ | 3↑1↓ | +2.9 | −5.0 |
| Estrogen Response Late | +0.2 | +0.8 | +3.4* | +7.0* | +6.3* | +4.2* | +3.4* | +6.9* | <.001 | 6↑0↓ | 3↑0↓ | −0.2 | −0.6 |
| Immunity & inflammation (category) | +2.3* | +3.0* | +3.9* | +4.3* | +3.6* | +3.6* | +2.8* | +7.2* | <.001 | 4↑0↓ | 6↑0↓ | −8.3 | +7.0 |
| Apoptosis | +1.5 | +2.7* | +4.5* | +4.0* | +6.3* | +2.1 | +2.8* | +4.9* | <.001 | 6↑0↓ | 4↑0↓ | −0.9 | +1.2 |
| Protein kinases (class) | +2.9* | +5.0* | +3.2* | +2.1 | +5.2* | +0.3 | +1.2 | +8.3* | <.001 | 3↑0↓ | 6↑0↓ | +7.0 | −7.4 |
| IL-6/JAK/STAT3 Signaling | +2.6* | +1.2 | +3.9* | +4.0* | +4.4* | +2.7* | +1.5 | +5.0* | <.001 | 6↑0↓ | 6↑0↓ | −4.8 | +3.7 |
| Homeodomain TFs (class) | +1.8 | +2.5* | −0.3 | +4.5* | +5.4* | +2.0 | +5.8* | +3.0* | <.001 | 5↑0↓ | 6↑0↓ | +3.1 | −2.7 |
| Interferon Gamma Response | +1.7 | +2.1 | +4.3* | +3.4* | +5.3* | +1.8 | +0.4 | +5.6* | <.001 | 4↑0↓ | 4↑0↓ | −6.6 | +5.3 |
| Androgen Response | +2.2 | +1.8 | +2.2* | +4.6* | +3.2* | +1.1 | +4.6* | +4.0* | <.001 | 4↑0↓ | 0↑0↓ | +0.4 | −0.6 |
| Glycolysis | +2.6* | +1.3 | +2.1 | +4.3* | +3.3* | +3.4* | +2.1 | +2.3* | <.001 | 5↑0↓ | 1↑0↓ | +1.6 | −1.2 |
| Xenobiotic Metabolism | −1.1 | −0.2 | +3.2* | +6.9* | +4.2* | +2.5* | +2.2* | +2.8* | <.001 | 5↑0↓ | 1↑0↓ | −1.1 | +0.2 |
| Hedgehog Signaling | | +2.5* | +1.5 | +2.8* | +4.0* | +2.1 | +3.0* | +3.1* | <.001 | 4↑0↓ | 1↑0↓ | +1.7 | −2.0 |
| Apical Surface | | +1.3 | +2.7* | +3.6* | +3.3* | | +3.6* | +3.0* | <.001 | 5↑0↓ | 2↑0↓ | −1.2 | +0.5 |
| Mitotic Spindle | +2.9* | +2.5* | +4.4* | −0.7 | +4.3* | +1.6 | +1.1 | +4.0* | <.001 | 2↑0↓ | 3↑0↓ | +1.1 | −2.0 |
| TGF-beta Signaling | +2.4* | +1.4 | +0.7 | +1.2 | +5.7* | +1.5 | +2.3* | +4.3* | <.001 | 3↑0↓ | 5↑0↓ | +2.0 | −3.3 |
| Nuclear receptors (class) | +1.3 | +4.7* | +0.6 | +5.9* | +0.8 | +1.9 | +0.6 | +3.6* | 0.002 | 2↑0↓ | 1↑0↓ | +3.0 | −2.8 |
| SLC transporters (class) | −0.3 | +5.3* | +2.0 | +8.8* | +1.0 | +1.3 | +1.1 | −0.2 | 0.002 | 2↑0↓ | 2↑0↓ | +0.5 | +0.5 |
| KRAS Signaling Dn | +1.3 | +2.2 | +1.6 | +5.6* | +0.1 | +1.4 | +3.1* | +2.4* | <.001 | 4↑0↓ | 3↑0↓ | +0.2 | −1.2 |
| p53 Pathway | +1.9 | −0.6 | +2.9* | +1.9 | +4.2* | +0.3 | +2.4* | +4.4* | <.001 | 5↑0↓ | 1↑0↓ | +0.2 | +0.1 |
| Cholesterol Homeostasis | −0.5 | +0.9 | +1.1 | +2.5* | +2.7* | +3.2* | +2.7* | +1.9 | <.001 | 4↑0↓ | 0↑2↓ | −1.8 | +1.3 |
| Transcription factors (class) | +5.9* | +3.9* | −2.2* | −0.6 | +1.0 | +0.4 | +1.1 | +4.4* | 0.002 | 3↑0↓ | 5↑0↓ | +1.4 | −1.4 |
| Notch Signaling | | −0.1 | +1.1 | +1.5 | +3.0* | +2.7* | +0.9 | +2.4* | 0.002 | 2↑0↓ | 0↑0↓ | +2.7 | −4.5 |
| mTORC1 Signaling | −1.7 | −1.4 | +5.6* | +2.3* | +3.8* | −0.1 | +0.6 | 0.0 | 0.002 | 2↑0↓ | 1↑1↓ | +3.4 | −2.7 |
| UV Response Up | −0.3 | −1.4 | +2.7* | +2.6* | +1.6 | +0.5 | +0.3 | +2.1* | 0.002 | 2↑0↓ | 0↑0↓ | +1.5 | −1.8 |
| **Recurrent, below the prior** | | | | | | | | | | | | | |
| G2-M Checkpoint | +1.0 | +0.1 | +7.2* | −6.1* | +0.5 | −1.2 | −4.1* | −3.2* | 0.002 | 1↑1↓ | 1↑1↓ | +3.0 | −2.4 |
| C2H2 zinc-finger TFs (class) | +3.2* | +0.1 | −4.2* | −3.2* | −4.4* | −1.3 | −2.9* | +0.6 | <.001 | 0↑4↓ | 0↑1↓ | −1.9 | +1.0 |
| Unfolded Protein Response | −0.7 | −5.3* | +2.7* | −3.4* | +1.9 | −2.2 | −3.3* | −3.0* | <.001 | 2↑2↓ | 2↑3↓ | +3.6 | −2.5 |
| Myc Targets V2 | −0.9 | −4.6* | +2.5* | −2.9* | −0.5 | −1.7 | −1.8 | −3.8* | 0.002 | 1↑1↓ | 1↑2↓ | −1.7 | +1.7 |
| E2F Targets | +0.1 | −3.5* | +9.0* | −6.4* | −2.7* | −4.0* | −3.4* | −4.1* | <.001 | 1↑4↓ | 1↑3↓ | +1.9 | −1.7 |
| Mitochondria & OXPHOS (category) | −3.2* | −3.1* | −4.7* | +1.7 | −3.1* | +1.2 | −5.0* | −7.6* | <.001 | 1↑6↓ | 0↑3↓ | −5.0 | +3.1 |
| DNA Repair | 0.0 | −6.7* | −0.1 | −2.8* | −3.6* | −2.2 | −4.5* | −4.8* | <.001 | 0↑5↓ | 0↑4↓ | +2.2 | −3.2 |
| Myc Targets V1 | −0.6 | −7.2* | +4.3* | −4.4* | −2.3* | −4.5* | −5.0* | −5.0* | <.001 | 1↑3↓ | 2↑4↓ | +7.0 | −6.6 |
| Protein homeostasis (category) | −1.9 | −4.7* | −1.8 | −3.4* | −1.6 | −5.5* | −5.8* | −2.7* | <.001 | 0↑4↓ | 0↑4↓ | +4.1 | −3.8 |
| Oxidative Phosphorylation | −2.6* | −9.3* | −3.1* | +1.6 | −3.0* | +0.7 | −5.3* | −9.4* | <.001 | 1↑5↓ | 0↑4↓ | −1.7 | +2.7 |
| Enzymes (class) | −6.2* | −2.0 | +0.2 | −4.7* | −7.3* | −1.4 | −5.3* | −4.8* | <.001 | 0↑5↓ | 0↑6↓ | +5.0 | −2.8 |
| Translation & ribosome (category) | +0.4 | −15.8* | 0.0 | +2.0 | −3.9* | −5.6* | −4.2* | −11.2* | <.001 | 2↑4↓ | 1↑3↓ | −0.5 | +1.8 |
| Chromatin regulators (class) | −3.7* | −1.1 | −3.4* | −12.0* | −5.5* | −4.6* | −6.5* | −3.4* | <.001 | 0↑7↓ | 1↑6↓ | +5.3 | −4.9 |
| Chromatin & DNA processing (category) | −1.7 | −7.2* | −0.8 | −9.0* | −6.5* | −5.7* | −8.7* | −8.8* | <.001 | 0↑5↓ | 0↑5↓ | +4.3 | −4.1 |
| Mitochondrial proteins (class) | −5.0* | −14.2* | −6.0* | −0.4 | −10.6* | −2.9* | −8.8* | −21.0* | <.001 | 0↑7↓ | 0↑6↓ | −8.3 | +9.0 |
| **Called in one tissue only** | | | | | | | | | | | | | |
| Heme Metabolism | −1.5 | −1.4 | +8.2* | +0.3 | +0.5 | +1.1 | +0.4 | +0.1 | 0.351 | 1↑2↓ | 0↑1↓ | +0.7 | +0.4 |
| Bile Acid Metabolism | +0.1 | +1.0 | +1.4 | +2.6* | −0.9 | +0.1 | −1.5 | −0.5 | 0.351 | 1↑1↓ | 1↑0↓ | −2.5 | +2.2 |
| Protein Secretion | +0.4 | −2.5* | +0.8 | −0.6 | +0.7 | −0.5 | −0.8 | −1.2 | 0.351 | 0↑0↓ | 0↑0↓ | +2.7 | −2.8 |

set $z$ (color clipped at ±8)

Cells: $z$ of the set against matched genes of the tissue; * $q < 0.05$; gray cell, fewer than 15 genes.
Right: $q$, recurrence across tissues; $S_g$ and profile correlation, tissues with the set above (↑) and below (↓); π and dN/dS, mean $z$ of the set over tissues. Gray type: not recurrent, or $|\text{mean } z|$ below 1.96.